

Assignment: Music Listening Logger (C Programming)

Problem Statement

Develop a console-based application called **Music Listening Logger** that allows users to:

1. Log music listening minutes for each day of the week.
2. Save the data to a file named music_log.txt.
3. View a weekly report showing total, average, and highest listening minutes.
4. Reset all data after confirmation.
5. Use a menu-driven interface until the user chooses to exit.

Complete C Program

```
#include <stdio.h>
```

```
int musicMinutes[7] = {0};
```

```
/* Function to save data to file */
```

```
void saveToFile()
```

```
{
```

```
    FILE *fp = fopen("music_log.txt", "w");
```

```
    for(int i = 0; i < 7; i++)
```

```
    {
```

```
        fprintf(fp, "%d\n", musicMinutes[i]);
```

```
    }
```

```
    fclose(fp);
```

```
}
```

```
/* Function to load data from file */  
void loadFromFile()  
{  
    FILE *fp = fopen("music_log.txt", "r");  
  
    if(fp == NULL)  
    {  
        return;  
    }  
  
    for(int i = 0; i < 7; i++)  
    {  
        fscanf(fp, "%d", &musicMinutes[i]);  
    }  
  
    fclose(fp);  
}
```

```
/* Function to log listening minutes */  
void logMinutes()  
{  
    printf("\nEnter listening minutes for 7 days:\n");  
  
    for(int i = 0; i < 7; i++)  
    {
```

```
        printf("Day %d: ", i + 1);

        scanf("%d", &musicMinutes[i]);
    }

    saveToFile();

    printf("\nData saved successfully.\n");
}

/* Function to generate report */
void weeklyReport()
{
    int total = 0;
    int highest = musicMinutes[0];

    for(int i = 0; i < 7; i++)
    {
        total += musicMinutes[i];

        if(musicMinutes[i] > highest)
        {
            highest = musicMinutes[i];
        }
    }

    float average = total / 7.0;
```

```

printf("\n===== Weekly Music Report =====\n");
printf("Total Listening Minutes : %d\n", total);
printf("Average Listening Time : %.2f\n", average);
printf("Highest Listening Time : %d\n", highest);
}

/* Function to display daily data */
void displayData()
{
    printf("\nListening Minutes:\n");

    for(int i = 0; i < 7; i++)
    {
        printf("Day %d : %d minutes\n",
            i + 1,
            musicMinutes[i]);
    }
}

/* Function to reset data */
void resetData()
{
    char choice;

    printf("\nAre you sure you want to delete all data? (Y/N): ");

```

```
scanf(" %c", &choice);
```

```
if(choice == 'Y' || choice == 'y')
```

```
{
```

```
    for(int i = 0; i < 7; i++)
```

```
    {
```

```
        musicMinutes[i] = 0;
```

```
    }
```

```
    FILE *fp = fopen("music_log.txt", "w");
```

```
    fclose(fp);
```

```
    printf("All data has been deleted.\n");
```

```
}
```

```
else
```

```
{
```

```
    printf("Reset cancelled.\n");
```

```
}
```

```
}
```

```
/* Main Function */
```

```
int main()
```

```
{
```

```
    int option;
```

```
    loadFromFile();
```

```
do
{
    printf("\n");
    printf("==== Music Listening Logger =====\n");
    printf("1. Log Listening Minutes\n");
    printf("2. View Weekly Data\n");
    printf("3. Generate Weekly Report\n");
    printf("4. Reset Weekly Data\n");
    printf("5. Exit\n");
    printf("Enter Choice: ");
    scanf("%d", &option);

    switch(option)
    {
        case 1:
            logMinutes();
            break;

        case 2:
            displayData();
            break;

        case 3:
            weeklyReport();
            break;
```

```
        case 4:
            resetData();
            break;

        case 5:
            printf("Exiting Application...\n");
            break;

        default:
            printf("Invalid Choice.\n");
    }

    } while(option != 5);

    return 0;
}
```

Sample Run

===== Music Listening Logger =====

1. Log Listening Minutes
2. View Weekly Data
3. Generate Weekly Report
4. Reset Weekly Data
5. Exit

Enter Choice: 1

Day 1: 60

Day 2: 45

Day 3: 90

Day 4: 120

Day 5: 50

Day 6: 75

Day 7: 100

Data saved successfully.

Weekly Report Example

===== Weekly Music Report =====

Total Listening Minutes : 540

Average Listening Time : 77.14

Highest Listening Time : 120

Features Implemented

- ✓ Array to store 7 days of listening minutes
 - ✓ Menu-driven application using loops and switch-case
 - ✓ File handling using music_log.txt
 - ✓ Weekly report generation
 - ✓ Calculation of total, average, and highest listening time
 - ✓ Data persistence through file storage
 - ✓ Reset option with confirmation message
 - ✓ Proper modular programming using functions
-

Conclusion

The Music Listening Logger demonstrates the use of arrays, functions, loops, file handling, menu-driven programming, and data analysis in C. The application allows users to store, retrieve, analyze, and manage their weekly music listening data efficiently.